

Moment-Fiber Stochastic Interpolants: Information-Projected Transport Between Observationally Equivalent Laws

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Abstract

Scientific inverse problems often determine only a finite set of population statistics rather than a unique microscopic law. We study the generative problem induced by this non-identifiability: how can one transport between distinct probability laws that agree on the measured expectations while preserving those expectations throughout the transport? We introduce *Moment-Fiber Stochastic Interpolants* (MFSI). Starting from a reference stochastic interpolant with marginal law $\tilde{Q}_t$, MFSI defines the target marginal at each time as the relative-entropy projection of $\tilde{Q}_t$ onto the population moment fiber $\{Q : \mathbb{E}_Q[\Phi] = \mathbf{c}\}$. The projected path is therefore a law-level object: it preserves the measured moments while making the smallest pointwise KL correction to the chosen reference path. We derive the exponential-tilt form of the projected marginals, their calibration sensitivities and scores, and an exact dynamic realization. In particular, the minimum- L^2 correction to a reference velocity is the gradient of a scalar potential solving a density-weighted Poisson equation with a forcing term that is computable from the moment multiplier, the reference velocity, and Jacobian-vector products of the measured observables. This yields a practical differentiable pipeline based on self-normalized importance weights, a low-dimensional convex calibration solve, and a Deep-Ritz correction network. The method is designed for *vertical transport* inside an observational equivalence class, rather than maximum-entropy generation from moments. We give exact and low-dimensional validation problems and a many-body benchmark in which pair statistics remain fixed while higher-order angular structure changes. Experiments are left as explicit placeholders in this draft.

1 Introduction

Many scientific measurements are reduced descriptions of a population. Examples include a finite set of correlation coefficients, scattering or structure-factor values, radial basis coefficients, empirical moments, low-frequency observables, and summary statistics of a field. Even when these measurements are accurate, they generally constrain a probability law without identifying it. The relevant inverse object is therefore not a single reconstruction but an *equivalence class of laws* consistent with the observation.

This paper asks a generative question inside that equivalence class. Suppose two scientifically distinct laws Q^- and Q^+ have the same measured expectations,

$$\mathbb{E}_{Q^-}[\Phi] = \mathbb{E}_{Q^+}[\Phi] = \mathbf{c}. \quad (1)$$

Can we construct a continuous generative transport from Q^- to Q^+ whose intermediate *population laws* remain compatible with the same observation? The requirement is deliberately law-level. We do not require every individual sample to satisfy $\Phi(x) = \mathbf{c}$; rather, we require

$$\mathbb{E}_{Q_t}[\Phi] = \mathbf{c}, \quad t \in [0, 1]. \quad (2)$$

This distinction matters in statistical mechanics and other ensemble sciences, where the measured object is often an expectation over configurations rather than a hard constraint on one configuration.

A generic stochastic interpolant between compatible endpoint distributions need not satisfy the same moment constraints at intermediate times, especially when Φ is nonlinear. Our proposal is to retain the reference interpolant as a transport geometry but correct its *marginal law* at every time by an information projection. This produces a canonical target path in probability space. The remaining technical question is then essential: how should one realize these projected marginals by an ODE or SDE? Merely correcting a velocity so that the desired moments

have the correct time derivative is insufficient, because infinitely many laws share the same finite set of moments. We therefore derive the full continuity-equation correction required to realize the projected density path.

The main contributions of the paper are:

1. **A law-level fiber path.** We define MFSI by pointwise relative-entropy projection of the marginals of a reference stochastic interpolant onto a population moment fiber. The construction has endpoint exactness and a simple integrated KL optimality property.
2. **Exact local calculus.** We derive the time and parameter sensitivity of the moment multiplier, the additive score correction, and the correct envelope derivative of the projection distortion. In particular, the distortion derivative itself does *not* require differentiating the optimal multiplier.
3. **Exact dynamic realization.** We show that the minimum-energy modification of a reference velocity that realizes the projected marginal path is a potential correction $-\nabla\psi_t$, where ψ_t solves a density-weighted Poisson equation. We derive a forcing formula that avoids explicit evaluation of $\partial_t \log \tilde{q}_t$ or a velocity divergence.
4. **A differentiable finite-particle algorithm.** The projected expectations are estimated by self-normalized exponential weights on ordinary stochastic-interpolant samples. The multiplier is found by a low-dimensional convex solve whose Jacobian is the weighted moment covariance. The Poisson correction is learned with a Deep-Ritz objective. Implicit differentiation is available when optimizing an upstream coupling or schedule, but is optional for the basic method.
5. **An empirical target with scientific content.** We propose a many-body test in which endpoint populations have the same smooth pair statistics but different angular or triplet organization. Success means transporting between these hidden structures while pair observables remain fixed on independent ensembles across the entire path.

The closest contemporary method is Moment Guided Diffusion (MGD), which also uses stochastic-interpolant ideas and population moments [1]. The distinction is substantive: MGD constructs dynamics that reproduce a finite moment trajectory and, in a large-volatility regime, target a maximum-entropy distribution. MFSI instead specifies a *full target marginal law* at every time by I-projection relative to a moving reference marginal and transports between two distinct endpoint laws that are already observationally equivalent. Moment preservation is necessary here but is not the defining objective.

2 Problem setup

2.1 Population moment fibers

Let $X \in \mathcal{X}$ denote a microscopic state and let $\Phi : \mathcal{X} \rightarrow \mathbb{R}^R$ be the measured observables. For a target vector $\mathbf{c} \in \mathbb{R}^R$, define the population moment fiber

$$\mathcal{F}(\mathbf{c}) = \{Q : \mathbb{E}_Q[\Phi(X)] = \mathbf{c}\}, \quad (3)$$

with absolute-continuity restrictions added when a physical prior or a hard state-space support requires them. The map $Q \mapsto \mathbb{E}_Q[\Phi]$ is many-to-one whenever the finite observation vector does not determine the law.

We assume access to endpoint distributions $Q^-, Q^+ \in \mathcal{F}(\mathbf{c})$. Their construction is modular. They may be empirical populations, distributions produced by two physical parameter regimes subsequently calibrated to the same observed moments, or compensated exponential tilts of a common physical prior. The endpoint mechanism is not the core contribution of this paper; the generative task begins once two compatible endpoint laws are available.

Scientific interpretation. A path within $\mathcal{F}(\mathbf{c})$ is a vertical motion in probability space: the microscopic population changes while the measured coordinates remain fixed. A path that changes $\mathbf{c}$ is horizontal. The present paper focuses on vertical transport.

2.2 Reference stochastic interpolant

Let (X^-, X^+, Z) have a chosen coupling with marginals $X^- \sim Q^-$ and $X^+ \sim Q^+$ and auxiliary noise Z . A stochastic interpolant is a differentiable map

$$\tilde{X}_t = I_t(X^-, X^+, Z), \quad t \in [0, 1], \quad (4)$$

with $\tilde{X}_0 = X^-$ and $\tilde{X}_1 = X^+$. Let $\tilde{Q}_t$ and $\tilde{q}_t$ denote its marginal law and density. Under the usual regularity conditions, the marginal density satisfies

$$\partial_t \tilde{q}_t + \nabla \cdot (\tilde{q}_t u_t) = 0, \quad (5)$$

where the reference velocity is the conditional expectation

$$u_t(x) = \mathbb{E}[\partial_t I_t(X^-, X^+, Z) \mid \tilde{X}_t = x], \quad (6)$$

which can be learned by the standard stochastic-interpolant or flow-matching regression objective [2, 3].

Although both endpoints lie on $\mathcal{F}(\mathbf{c})$, $\tilde{Q}_t$ generally does not. For nonlinear Φ , even a simple Euclidean interpolation between endpoint samples can change low-order population moments substantially.

3 Information-projected stochastic-interpolant marginals

3.1 Definition and exponential form

For each time t , define the projected marginal

$$Q_t^{\text{fib}} \in \arg \min_{Q: \mathbb{E}_Q[\Phi] = \mathbf{c}_t} \text{KL}(Q \parallel \tilde{Q}_t), \quad (7)$$

where $\mathbf{c}_t \equiv \mathbf{c}$ for the vertical transport studied here. We retain $\mathbf{c}_t$ in the derivations because it clarifies the calculus and allows later extension to changing observations.

Define

$$A_t(\boldsymbol{\lambda}) = \log \mathbb{E}_{\tilde{Q}_t} \left[e^{\boldsymbol{\lambda}^\top \Phi(X)} \right]. \quad (8)$$

Whenever the dual solution exists, the I-projection has density

$$q_t(x) = \tilde{q}_t(x) \exp(\boldsymbol{\lambda}_t^\top \Phi(x) - A_t(\boldsymbol{\lambda}_t)), \quad \nabla_{\boldsymbol{\lambda}} A_t(\boldsymbol{\lambda}_t) = \mathbf{c}_t, \quad (9)$$

where, to simplify notation, q_t denotes the density of Q_t^{fib} below.

Assumption 1 (Regular projected path). For every $t \in [0, 1]$, the log moment-generating function A_t is finite in a neighborhood of a dual optimizer $\boldsymbol{\lambda}_t$, the target $\mathbf{c}_t$ lies in the relative interior of the moment set induced by $\tilde{Q}_t$, and the covariance

$$C_t = \text{Cov}_{Q_t^{\text{fib}}}(\Phi, \Phi) \quad (10)$$

is positive definite on the identifiable span of the measured observables. In implementation, redundant directions are removed or treated with a rank-aware pseudoinverse.

Proposition 1 (Endpoint exactness and pathwise KL optimality). *Under the regular projected-path assumption, $\mathbb{E}_{Q_t^{\text{fib}}}[\Phi] = \mathbf{c}_t$ for every t . If $\tilde{Q}_0 = Q^-$ and $\tilde{Q}_1 = Q^+$ with $Q^-, Q^+ \in \mathcal{F}(\mathbf{c})$, then $Q_0^{\text{fib}} = Q^-$ and $Q_1^{\text{fib}} = Q^+$. Moreover, for every nonnegative integrable weight $w(t)$ and every feasible law path R_t satisfying $\mathbb{E}_{R_t}[\Phi] = \mathbf{c}_t$,*

$$\int_0^1 w(t) \text{KL}(R_t \parallel \tilde{Q}_t) dt \geq \int_0^1 w(t) \text{KL}(Q_t^{\text{fib}} \parallel \tilde{Q}_t) dt. \quad (11)$$

The statement is intentionally modest: it is a pointwise I-projection applied over time, not a path-space entropy optimum. Its value is that it defines a precise *full marginal target* for within-fiber generation.

3.2 Sensitivity and score identities

Let

$$\tilde{\ell}_t(x) = \partial_t \log \tilde{q}_t(x). \quad (12)$$

Differentiating the calibration equation in [equation \(9\)](#) gives

Proposition 2 (Multiplier sensitivity). *Under differentiation-under-the-integral regularity,*

$$C_t \dot{\boldsymbol{\lambda}}_t + \text{Cov}_{Q_t^{\text{fib}}}(\Phi, \tilde{\ell}_t) - \dot{\mathbf{c}}_t = 0, \quad (13)$$

so that

$$\dot{\boldsymbol{\lambda}}_t = C_t^{-1} [\dot{\mathbf{c}}_t - \text{Cov}_{Q_t^{\text{fib}}}(\Phi, \tilde{\ell}_t)]. \quad (14)$$

The same identity holds for a differentiable parameter ϑ after replacing time derivatives by ∂_ϑ and including $\partial_\vartheta \mathbf{c}_t$.

If Φ is differentiable in the state variable, then the projected score is

$$\nabla_x \log q_t(x) = \nabla_x \log \tilde{q}_t(x) + J_\Phi(x)^\top \lambda_t. \quad (15)$$

Thus a score model for the reference interpolant can be converted into a score model for the projected path by a low-dimensional scientific correction.

3.3 Projection distortion and the envelope correction

Define the instantaneous projection distortion

$$D_t = \text{KL}(Q_t^{\text{fib}} \parallel \tilde{Q}_t) = \lambda_t^\top \mathbf{c}_t - A_t(\lambda_t) \geq 0. \quad (16)$$

This quantity measures how far the chosen reference marginal cuts away from the moment fiber.

A useful correction to an earlier formulation is that differentiating D_t does *not* require differentiating the optimal multiplier. The dual first-order condition produces an envelope identity.

Proposition 3 (Envelope derivative of projection distortion). *For any scalar parameter ϑ on which $\tilde{q}_t$ and possibly $\mathbf{c}_t$ depend,*

$$\partial_\vartheta D_t = \lambda_t^\top \partial_\vartheta \mathbf{c}_t - \mathbb{E}_{Q_t^{\text{fib}}}[\partial_\vartheta \log \tilde{q}_t(X)]. \quad (17)$$

In particular, the terms proportional to $\partial_\vartheta \lambda_t$ cancel.

This matters for how differentiability is advertised. Implicit differentiation through the multiplier remains useful for objectives that depend on λ_t , $\tilde{\lambda}_t$, the projected score, or the dynamically corrected sampler, but it is not mathematically required merely to differentiate the optimum value D_t .

4 From projected marginals to an exact transport

4.1 Why moment-tangent projection is not enough

Suppose one modifies a candidate velocity u_t only so that

$$\frac{d}{dt} \mathbb{E}[\Phi(X_t)] = \dot{\mathbf{c}}_t. \quad (18)$$

Such a correction constrains only R scalar quantities. It does not identify the complete density evolution q_t and therefore does not, in general, realize the I-projected marginals in [equation \(9\)](#). This distinction is central to the method: *moment preservation is weaker than marginal-path realization*.

The desired velocity must satisfy the full continuity equation

$$\partial_t q_t + \nabla \cdot (q_t v_t) = 0. \quad (19)$$

Rather than discard the reference velocity u_t , we seek the smallest $L^2(q_t)$ correction that makes [equation \(19\)](#) exact.

4.2 Minimal kinetic correction

Define the continuity residual of the reference velocity under the projected law,

$$h_t(x) := \frac{\partial_t q_t(x) + \nabla \cdot (q_t(x) u_t(x))}{q_t(x)}. \quad (20)$$

By mass conservation, $\mathbb{E}_{q_t}[h_t] = 0$. Consider the weighted Poisson problem

$$\nabla \cdot (q_t \nabla \psi_t) = q_t h_t, \quad \mathbb{E}_{q_t}[\psi_t] = 0, \quad (21)$$

where the mean-zero condition fixes the additive gauge.

Theorem 4 (Exact minimal-energy realization). *Assume q_t is positive and sufficiently regular, $h_t \in L^2(q_t)$ has zero mean, and the weighted Poisson problem [equation \(21\)](#) admits a weak solution with $\nabla\psi_t \in L^2(q_t)$. Define*

$$v_t^*(x) = u_t(x) - \nabla\psi_t(x). \quad (22)$$

Then v_t^ realizes the projected marginal path, i.e. it satisfies [equation \(19\)](#). Moreover, among all finite-energy velocities v_t satisfying [equation \(19\)](#), v_t^* minimizes*

$$\mathbb{E}_{q_t} \left[\|v_t - u_t\|_2^2 \right]. \quad (23)$$

The minimizer is unique in $L^2(q_t)$ modulo q_t -null sets.

Intuition. The reference interpolant proposes how probability should move. The I-projection says where the probability distribution is allowed to be. The Poisson field supplies only the additional mass rearrangement needed to reconcile those two statements. It is the distribution-level analogue of an orthogonal projection: not onto a sample manifold, but onto the tangent equation of a prescribed probability-density path.

4.3 A forcing formula that avoids density time derivatives

Directly evaluating [equation \(20\)](#) appears unattractive because it contains $\partial_t q_t$, a divergence, and the unknown reference density. The exponential structure makes the residual much simpler.

Let

$$m_t(x) = J_{\Phi}(x)u_t(x) \in \mathbb{R}^R. \quad (24)$$

The quantity $m_t(x)$ is a Jacobian-vector product: the instantaneous rate at which the reference velocity changes the measured statistics at state x .

Proposition 5 (Computable multiplier derivative and Poisson forcing). *Assume the reference velocity satisfies [equation \(5\)](#). Then*

$$\text{Cov}_{q_t}(\Phi, \tilde{\ell}_t) = \mathbb{E}_{q_t}[m_t] + \mathbb{E}_{q_t}[(\Phi - \mathbf{c}_t)(\lambda_t^\top m_t)], \quad (25)$$

so the multiplier derivative can be computed without $\tilde{\ell}_t$ as

$$C_t \dot{\lambda}_t = \dot{\mathbf{c}}_t - \mathbb{E}_{q_t}[m_t] - \mathbb{E}_{q_t}[(\Phi - \mathbf{c}_t)(\lambda_t^\top m_t)]. \quad (26)$$

Furthermore, the continuity residual is

$$h_t(x) = \dot{\lambda}_t^\top (\Phi(x) - \mathbf{c}_t) + \lambda_t^\top (m_t(x) - \mathbb{E}_{q_t}[m_t]). \quad (27)$$

Hence h_t requires only λ_t , one $R \times R$ covariance solve, evaluations of u_t , and Jacobian-vector products through Φ .

For the vertical problem, $\dot{\mathbf{c}}_t = 0$. No score network is required for the deterministic implementation.

4.4 Deep-Ritz training of the correction

[Equation \(21\)](#) is the Euler–Lagrange equation of the Dirichlet functional

$$\mathcal{J}_t(\psi) = \frac{1}{2} \mathbb{E}_{q_t} [\|\nabla\psi(X)\|_2^2] + \mathbb{E}_{q_t} [h_t(X)\psi(X)]. \quad (28)$$

Therefore a scalar neural potential $\psi_\omega(t, x)$ can be trained by minimizing

$$\mathcal{L}_{\text{Ritz}}(\omega) = \mathbb{E}_{t \sim \rho} [\mathcal{J}_t(\psi_\omega(t, \cdot))]. \quad (29)$$

The objective uses only samples from q_t and first derivatives of the scalar network. Related density-weighted Poisson and Deep-Ritz constructions appear in filtering and potential-flow generative modeling [4, 5]; we do not claim the Poisson solver as an isolated novelty. The contribution here is the specific I-projected stochastic-interpolant target and the closed-form forcing induced by the moment calibration.

Corollary 6 (Optional stochastic sampler). *If a reference score $\tilde{s}_t = \nabla \log \tilde{q}_t$ is available, define*

$$s_t(x) = \tilde{s}_t(x) + J_{\Phi}(x)^{\top} \lambda_t. \quad (30)$$

For any nonnegative volatility schedule ε_t , the SDE

$$dX_t = (v_t^*(X_t) + \varepsilon_t s_t(X_t)) dt + \sqrt{2\varepsilon_t} dW_t \quad (31)$$

has the same target marginals q_t as the deterministic flow, under the standard Fokker–Planck regularity assumptions.

For the hackathon, the deterministic ODE is the preferred first implementation; the SDE requires an additional reference-score model and is not necessary to demonstrate the paper thesis.

5 Finite-particle and differentiable implementation

5.1 Self-normalized I-projection on interpolant samples

A major practical advantage is that training does not require drawing fresh MCMC samples from every projected marginal. At a sampled time t , draw an ordinary stochastic-interpolant batch

$$x_i \sim \tilde{Q}_t, \quad i = 1, \dots, M. \quad (32)$$

For a candidate multiplier, set

$$a_i = \lambda^{\top} \Phi(x_i), \quad \bar{w}_i(\lambda) = \frac{e^{a_i - a_{\max}}}{\sum_j e^{a_j - a_{\max}}}. \quad (33)$$

The empirical dual objective is

$$\hat{f}_t(\lambda) = \log \left(\frac{1}{M} \sum_i e^{\lambda^{\top} \Phi(x_i)} \right) - \lambda^{\top} c_t, \quad (34)$$

with gradient and Hessian

$$\nabla \hat{f}_t(\lambda) = \sum_i \bar{w}_i \Phi(x_i) - c_t, \quad (35)$$

$$\nabla^2 \hat{f}_t(\lambda) = \widehat{\text{Cov}}_{\bar{w}}(\Phi, \Phi). \quad (36)$$

Thus Newton, damped Newton, or L-BFGS calibration is low dimensional even when the microscopic state dimension is large. Any expectation under q_t in equations (26) to (28) is then estimated by the same normalized weights.

The empirical effective sample size

$$\text{ESS}_t = \frac{1}{\sum_i \bar{w}_i^2} \quad (37)$$

is a first-class diagnostic. A large projection distortion with a small ESS_t means that the chosen reference interpolant has poor overlap with the fiber-corrected marginal, regardless of whether the low-dimensional multiplier solve converges numerically.

Finite-batch feasibility. The population projection may exist even when a particular minibatch cannot represent the target moment vector. Empirically, exact calibration requires c_t to lie in the relative interior of the convex hull of $\{\Phi(x_i)\}$. We therefore recommend a large time-local sample bank, common random numbers during the inner solve, damping, rank truncation, and an explicit failure flag when the empirical convex hull or ESS is inadequate. A soft ridge on the dual is acceptable as a numerical fallback but should be reported as approximate rather than exact projection.

5.2 Reference-velocity training

The reference velocity network $u_{\theta}(t, x)$ is trained conventionally:

$$\mathcal{L}_{\text{SI}}(\theta) = \mathbb{E} \left[\left\| u_{\theta}(t, \tilde{X}_t) - \partial_t I_t(X^-, X^+, Z) \right\|_2^2 \right]. \quad (38)$$

We recommend training u_{θ} first, freezing it for the initial MFSI implementation, and then training the scalar correction potential ψ_{ω} . Joint training is possible but is not required to establish the method.

5.3 Computing the correction without a full Jacobian

For every projected batch point, compute

$$m_i = J_{\Phi}(x_i)u_{\theta}(t, x_i) \quad (39)$$

with a Jacobian-vector product. There is no need to materialize the full $R \times d$ Jacobian. Weighted batch statistics give C_t , $\mathbb{E}[m_t]$, and the covariance term in [equation \(26\)](#). One $R \times R$ linear solve yields $\dot{\lambda}_t$, after which [equation \(27\)](#) gives the scalar forcing h_i for every sample.

The correction network is a scalar potential. For particles, making ψ_{ω} invariant under particle permutations and Euclidean symmetries automatically makes $-\nabla_X \psi_{\omega}$ an equivariant velocity correction. This is attractive both physically and computationally.

5.4 Where implicit differentiation is actually useful

Let the empirical calibration equation be

$$F(\lambda, \xi) = \widehat{\mathbb{E}}_{q_{\lambda, \xi}}[\Phi] - c(\xi) = 0, \quad (40)$$

where ξ denotes any upstream differentiable parameter, such as an interpolant schedule, coupling temperature, physical parameter, or learned endpoint parameter. On the identifiable span,

$$\frac{d\lambda^*}{d\xi} = -C^{-1} \partial_{\xi} F. \quad (41)$$

For a reverse-mode cotangent $\bar{\lambda}$, the backward pass solves

$$C^{\top} a = \bar{\lambda} \quad (42)$$

and contributes $-(\partial_{\xi} F)^{\top} a$ upstream. This avoids unrolling all Newton iterations.

We propose a hierarchy rather than making end-to-end differentiability mandatory:

Level 0: core method. Train u_{θ} , solve λ_t numerically with stop-gradient, compute $\dot{\lambda}_t$ from [equation \(26\)](#), and train ψ_{ω} . This already realizes the paper’s generative construction.

Level 1: differentiable calibration. Add a custom implicit VJP for the multiplier and verify it against finite differences on toy problems.

Level 2: fiber-adapted reference path. Parameterize a coupling or schedule by ξ and reduce the amount of dynamic correction required, for example through the integrated correction energy

$$\mathcal{E}_{\text{corr}}(\xi) = \int_0^1 \mathbb{E}_{q_t^{\xi}} \left[\left\| \nabla \psi_t^{\xi} \right\|_2^2 \right] dt, \quad (43)$$

subject to an ESS penalty. Here implicit differentiation through the calibration is useful and scientifically interpretable: it learns a reference transport that naturally stays closer to the observational equivalence class.

We do *not* use projection distortion alone as the flagship differentiable objective, because its optimum-value derivative obeys the envelope identity [equation \(17\)](#); claiming that it intrinsically requires $d\lambda/d\xi$ would be incorrect.

5.5 Optional population-moment safety layer

At inference time, an interacting ensemble can be given an additional low-rank correction that eliminates empirical moment-rate drift. For an ensemble law Q_t and candidate velocity v_t , define

$$G_t = \mathbb{E}[J_{\Phi} J_{\Phi}^{\top}], \quad r_t = \mathbb{E}[J_{\Phi} v_t] - \dot{c}_t. \quad (44)$$

Then

$$v_t^{\text{safe}}(x) = v_t(x) - J_{\Phi}(x)^{\top} G_t^{\dagger} r_t \quad (45)$$

is the minimum- L^2 correction in the span of $J_{\Phi}^{\top}$ that fixes the desired moment rate, when reachable. This is an implementation safeguard, not the dynamic realization theorem: [equation \(45\)](#) alone does not force the full law to equal q_t . Closely related Gram-matrix moment guidance is already present in MGD [1].

5.6 Algorithm

1. **Endpoint data.** Obtain samples from two laws $Q^-, Q^+ \in \mathcal{F}(\mathbf{c})$ that differ in hidden structure.
2. **Reference bridge.** Choose a coupling and stochastic interpolant I_t ; train u_θ with equation (38).
3. **Projected minibatch.** For a sampled t , draw $x_i \sim \tilde{Q}_t$ and solve equation (34) for λ_t .
4. **Fiber calculus.** Use normalized weights to estimate C_t and projected expectations. Compute $m_i = J_\Phi(x_i)u_\theta(t, x_i)$, solve equation (26), and evaluate h_i from equation (27).
5. **Dynamic correction.** Update ω using the weighted empirical version of equation (29).
6. **Generation.** Sample $X_0 \sim Q^-$ and integrate

$$\dot{X}_t = u_\theta(t, X_t) - \nabla_X \psi_\omega(t, X_t). \quad (46)$$

Optionally apply equation (45) using the concurrently generated ensemble.

7. **Evaluation.** At held-out time points, compare generated samples against independently importance-resampled or quadrature-computed Q_t^{fib} and test moments on fresh ensembles.

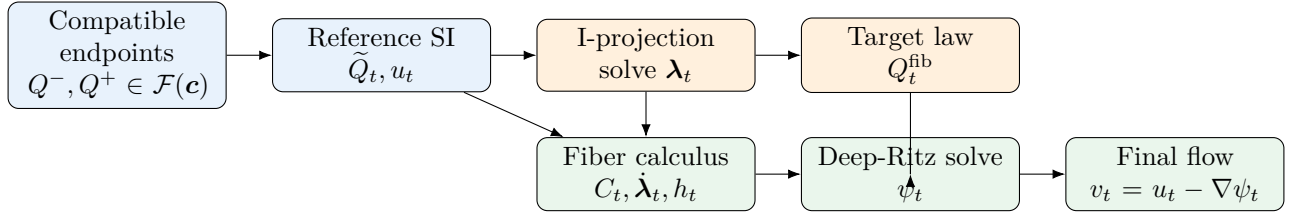


Figure 1: MFSI separates the target law path from its dynamic realization. The orange block is a low-dimensional information projection; the green block learns the minimum-energy velocity correction required to realize the projected path.

6 Empirical program

The empirical study should test more than moment preservation. A method can preserve the measured moments while following the wrong distributional path. We therefore evaluate three levels: (i) fiber fidelity, (ii) marginal-path fidelity, and (iii) useful hidden-structure transport.

6.1 Experiment A: one-dimensional exact reference

Let

$$Q^- = \mathcal{N}(0, 1), \quad Q^+ = \frac{1}{2}\mathcal{N}(-a, 1 - a^2) + \frac{1}{2}\mathcal{N}(a, 1 - a^2), \quad 0 < a < 1. \quad (47)$$

Both endpoints have mean zero and variance one. Set

$$\Phi(x) = (x, x^2), \quad H(x) = x^4. \quad (48)$$

Use a standard stochastic interpolant between endpoint samples. At each t , compute the projected density by high-accuracy quadrature and solve the one-dimensional weighted Poisson equation numerically. This gives a near-exact oracle for every component of the method.

The required checks are:

- the calibrated path maintains mean zero and second moment one to quadrature accuracy;
- the empirical multiplier, $\dot{\lambda}_t$, and h_t match the quadrature reference;
- the learned Deep-Ritz correction reproduces the oracle velocity and projected marginals;
- the fourth moment changes continuously between endpoint values while the measured moments remain fixed;
- implicit VJPs agree with central finite differences for a simple interpolant parameter.

[**RESULT TO INSERT: Insert moment-error, density-overlay, velocity-error, and gradient-check plots.**]

6.2 Experiment B: equal-covariance multimodality in two dimensions

Construct a ring-like law and a four-lobed law with identical mean and covariance. Use first and second moments as Φ and angular Fourier coefficients as held-out descriptors. This experiment tests whether MFSI controls the complete projected path rather than merely matching the same covariance.

Compare:

1. raw stochastic interpolant;
2. MGD-style moment guidance;
3. moment-tangent projection using equation (45) without the Poisson correction;
4. MFSI with the exact projected target and learned Poisson correction.

The decisive metric is not just moment error. At a held-out grid of times, draw an oracle sample from Q_t^{fib} by importance resampling from a large reference bank and compute a two-sample discrepancy (energy distance, MMD, or a carefully cross-validated classifier two-sample statistic) between oracle and generated particles. MGD and the tangent-only baseline may have small moment error while differing from the target projected law; MFSI should reduce this marginal discrepancy.

[RESULT TO INSERT: Insert path snapshots, moment error, two-sample distance, hidden angular coefficients, correction energy, and ESS.]

6.3 Experiment C: many-body fiber bridge

Physical system. Use $N \approx 32$ particles in a two-dimensional periodic square. A convenient reference family has pair and angular interactions,

$$U_\eta(X) = \sum_{i < j} U_2(r_{ij}; \eta_2) + \sum_{i, j, k} U_3(r_{ij}, r_{ik}, \angle jik; \eta_3), \quad (49)$$

with a soft repulsive core so the relevant state-space support is not artificially disconnected.

The measured observables are smooth radial pair coefficients

$$\Phi_a(X) = \frac{1}{N(N-1)} \sum_{i \neq j} \varphi_a(r_{ij}), \quad a = 1, \dots, R, \quad (50)$$

where φ_a are Gaussian radial basis functions. Held-out structure includes bond-orientational order, smooth triplet coefficients, local motif frequencies, and selected structure-factor values.

Compatible endpoints. Produce two endpoint populations with visibly different angular organization. The cleanest construction is to vary an angular interaction or hidden exponential field and recalibrate low-dimensional pair fields until the pair-observation vector equals a common target $\mathbf{c}$ within independent Monte Carlo uncertainty. This endpoint calibration is auxiliary; MFSI begins from the resulting samples.

Particle coupling and periodic interpolation. Particle labels are exchangeable. For each paired endpoint configuration, first align global translation, compute a periodic minimum-image assignment between particles using Hungarian matching or entropic Sinkhorn, and interpolate along wrapped displacements. Independent configuration pairing is the simplest baseline; minibatch optimal-transport pairing can be tested as a lower-distortion reference bridge. If gradients through the pairing are desired, use Sinkhorn; otherwise the pairing may be treated as preprocessing.

Networks. Use a permutation- and translation-equivariant periodic graph network for $u_\theta(t, X)$, built from minimum-image relative coordinates. Parameterize $\psi_\omega(t, X)$ as a scalar network invariant under the same symmetries; automatic differentiation then produces an equivariant correction force $-\nabla_X \psi_\omega$. If the state representation and box geometry are chosen to preserve continuous rotational symmetry, the corresponding rotation-invariant/equivariant architecture can be used as well. All measured pair statistics are differentiable, and $J_\Phi u$ is obtained by JVP.

Primary empirical claim. A convincing result has three simultaneous properties:

1. pair observables remain statistically indistinguishable from $\mathbf{c}$ on *fresh generated ensembles* over the complete path;
2. higher-order structure moves continuously from the Q^- regime to the Q^+ regime;
3. generated intermediate laws agree substantially better with independently estimated projected marginals than moment-only baselines do.

[RESULT TO INSERT: Insert many-body path visualization, radial observables, angular/triplet observables, two-sample projected-path fidelity, ESS, and wall-clock/NFE comparisons.]

6.4 Metrics

We recommend reporting the following quantities over a held-out time grid $\{t_j\}$:

$$\epsilon_{\Phi}(t) = \left\| \widehat{\mathbb{E}}[\Phi] - \mathbf{c} \right\|_W, \quad (51)$$

$$\epsilon_{\text{path}}(t) = d\left(\widehat{Q}_t^{\text{gen}}, \widehat{Q}_t^{\text{fib}}\right), \quad (52)$$

$$E_{\text{corr}} = \int_0^1 \widehat{\mathbb{E}}_{q_t} \|\nabla \psi_{\omega}\|_2^2 dt, \quad (53)$$

$$D_{\text{proj}} = \int_0^1 \widehat{D}_t dt, \quad (54)$$

where W may whiten moment components using independent uncertainty, and d is a distributional two-sample metric. Also report minimum/median ESS, endpoint fidelity, hidden-descriptor trajectories, ODE function evaluations, and wall-clock cost.

A useful scientific plot is a *fiber profile*: measured observables remain flat while one or more held-out descriptors move substantially. A useful methodological plot overlays this with path discrepancy to the I-projected oracle; this prevents a moment-only method from appearing equivalent to MFSI simply because both satisfy $\mathbb{E}[\Phi] = \mathbf{c}$.

6.5 Ablations and falsification tests

The method should be considered unsupported if any of the following occur systematically:

- projected importance weights collapse at most intermediate times, indicating that the reference bridge has insufficient overlap;
- the Deep-Ritz correction preserves moments but fails the full marginal two-sample test;
- a much cheaper moment-only baseline gives indistinguishable projected-path fidelity;
- the learned velocity reaches the correct endpoint only by violating pair moments in the interior;
- claimed gradient advantages disappear once matched-compute finite differences are used for the small number of optimized parameters.

Ablations should include correction-network capacity, batch size, covariance rank truncation, reference coupling, stochastic-interpolant noise level, safety-layer use, and stop-gradient versus implicit differentiation through calibration.

7 Related work and novelty boundary

Stochastic interpolants and flow matching. Stochastic interpolants and flow matching construct learnable finite-time transports between probability laws and provide regression objectives for the associated velocity or score fields [2, 3]. MFSI uses such a path only as a reference geometry; the target marginal path is replaced by its information projection onto a population expectation fiber.

Moment Guided Diffusion. MGD is the closest work conceptually [1]. It constructs a nonlinear diffusion whose finite set of moments follows the moments of a stochastic interpolant and studies convergence toward a maximum-entropy distribution. MGD explicitly emphasizes that its guided drift reproduces the prescribed moments rather than the full stochastic-interpolant distribution. MFSI addresses a different task: both endpoints are already

compatible with the same observation, and the full intermediate target law is specified by relative-entropy projection of each reference marginal. The comparison to MGD is therefore essential: matching the same moments is a baseline, not evidence that the projected marginal path has been realized.

Hard and sample-level constrained flows. Stiefel Flow Matching constrains each molecular point cloud to a manifold of fixed moments of inertia [6]. Physics-Constrained Flow Matching and HardFlow impose hard constraints on individual generated states or trajectories [7, 8]. These are different constraint objects. MFSI constrains an expectation under the generated population, and an individual sample generally does not satisfy $\Phi(x) = c$.

Information projection and path-space entropy. I-projection onto linear expectation constraints and its exponential-family form are classical [9]. Maximum Caliber applies entropy principles to distributions over trajectories under dynamical constraints [10]. Schrodinger-bridge methods instead minimize a path-space relative entropy subject to prescribed marginal distributions. MFSI uses classical I-projection differently: at every time it constrains only selected expectations and takes the moving stochastic-interpolant marginal as the reference measure. We should not claim novelty for exponential tilting itself.

Potential flows and weighted Poisson equations. Density-weighted Poisson equations appear in feedback-particle filtering and related McKean–Vlasov representations [4]. Recent potential-flow generative modeling also uses Deep-Ritz ideas to realize a prescribed density homotopy [5]. This makes the novelty boundary particularly clear: the weighted-Poisson realization is an enabling mechanism; the proposed object is a moment-fiber I-projected stochastic-interpolant path and the closed-form low-dimensional forcing that couples it to measured population observables.

Novelty hypothesis. Subject to a fuller literature review, the defensible methodological claim is the following combination:

Given two distinct endpoint laws that are observationally equivalent under population moments, define a canonical law-valued bridge by information-projecting a stochastic-interpolant marginal at every time onto the common moment fiber, and realize that complete projected path with the minimum-energy correction of the reference velocity.

The paper should not claim that moment preservation, exponential tilting, implicit calibration, or weighted Poisson transport is individually new.

8 Limitations

Overlap and weight degeneracy. MFSI depends on overlap between $\tilde{Q}_t$ and the projected target. If the I-projection requires an extreme tilt, self-normalized importance weights collapse. This is not merely a numerical inconvenience: it signals a poor reference path. Coupling design, schedule design, or sequential intermediate projections may be needed.

Finite-sample versus population constraints. Exact theoretical moment preservation is a statement about the population law. Finite generated ensembles fluctuate. The optional safety layer can control empirical moment drift but introduces particle interaction and additional cost. All reported moment errors should therefore be compared with independent sampling uncertainty.

Singular moment geometry. If measured statistics are redundant, C_t is singular. The theory should be interpreted on the identifiable quotient space. The implementation must use whitening, rank detection, and the same regularization in forward and backward linear solves.

Hard support constraints. An ordinary Euclidean interpolant may leave a physically admissible state space. For periodic particles with soft interactions this is manageable; for rigid manifolds or hard-core support, the reference interpolant itself must respect the state space. MFSI does not remove that requirement.

Poisson approximation. The exact dynamic theorem assumes the weighted Poisson equation is solved accurately. A finite neural potential introduces transport error. This is why the empirical study must compare generated intermediate laws with independently estimated projected marginals, not only with their moments.

No universal advantage from differentiation. The multiplier dimension R may be small enough that black-box optimization is already cheap. Differentiability is most valuable when optimizing high-dimensional upstream parameters, amortizing over many time points, or differentiating a scientifically meaningful path objective. The paper should report matched-compute comparisons rather than presume an advantage.

9 Discussion

MFSI changes the role of moment constraints in generative modeling. The moments are not used to define a maximum-entropy terminal distribution and are not hard constraints on individual samples. Instead they define an observational equivalence class, and the generative model is asked to move *within* that class.

The construction separates three objects that are easy to conflate. First, the reference stochastic interpolant supplies a convenient coupling geometry and a learnable base velocity. Second, the I-projection defines the desired law path exactly. Third, the weighted-Poisson correction realizes that target path dynamically. The separation is scientifically useful because each failure has a different interpretation: large D_t or low ESS indicates a poor reference path; ill-conditioned C_t indicates poorly identified or redundant measurements; large Ritz error indicates an inadequate dynamic correction model.

The differentiable pipeline is similarly modular. The basic method requires only ordinary automatic differentiation for observables and the scalar correction network. An implicit VJP through the low-dimensional calibration becomes useful when the bridge itself is optimized to reduce correction energy or when endpoint physics is learned jointly. This keeps the hackathon target realistic while preserving a credible path toward a differentiable scientific-computing contribution.

A particularly interesting extension is a conditional MFSI model over an ambiguity coordinate a : instead of learning one bridge $Q^- \rightarrow Q^+$, learn a family $Q_a \in \mathcal{F}(\mathbf{c})$ and a velocity $v(t, x, a_0, a_1)$ that transports between arbitrary compatible laws. Another extension is active measurement design: once MFSI exposes large hidden variation along a fiber, select a new observable that most strongly distinguishes the generated alternatives. These are natural follow-on projects, but they are not required for the core paper.

10 Conclusion

We proposed Moment-Fiber Stochastic Interpolants for generative transport between observationally equivalent probability laws. The defining object is a time-indexed relative-entropy projection of a reference stochastic-interpolant marginal onto a population moment fiber. This produces a complete target law at every time, not only a prescribed moment trajectory. We derived its exponential form, multiplier sensitivity, score, projection distortion, and the envelope derivative that corrects an earlier differentiability claim. We then closed the main dynamic gap: the exact minimum-energy modification of a reference velocity is a weighted-Poisson potential correction, with a forcing term computable from the calibrated low-dimensional field and Jacobian-vector products of the measured observables. The resulting implementation uses standard stochastic-interpolant samples, self-normalized weights, small covariance solves, and a Deep-Ritz network. The experimental program is designed to distinguish true projected-path fidelity from mere moment matching, culminating in a many-body bridge that changes angular or triplet organization while pair statistics remain fixed.

A Proofs

A.1 I-projection form and endpoint exactness

For fixed t , minimize $\text{KL}(Q \parallel \tilde{Q}_t)$ subject to normalization and $\mathbb{E}_Q[\Phi] = \mathbf{c}_t$. The standard convex dual introduces a multiplier λ and yields the exponential density ratio in equation (9). Strict convexity of KL gives uniqueness of the projected law whenever the feasible set is nonempty and the dual optimum is regular [9]. If $\tilde{Q}_t$ is already feasible, it achieves objective value zero and is therefore the unique projection. Applying this at $t = 0, 1$ proves endpoint exactness. The integrated inequality equation (11) follows by integrating the pointwise optimality inequality against $w(t) \geq 0$.

A.2 Multiplier sensitivity

The calibration equation is

$$\nabla_{\lambda} A_t(\lambda_t) = \mathbf{c}_t. \quad (55)$$

Its derivative is

$$\nabla_{\lambda}^2 A_t(\lambda_t) \dot{\lambda}_t + \partial_t \nabla_{\lambda} A_t(\lambda_t) = \dot{\mathbf{c}}_t. \quad (56)$$

The Hessian is C'_t . At fixed λ ,

$$\partial_t \nabla_{\lambda} A_t(\lambda) = \text{Cov}_{Q_t, \lambda}(\Phi, \partial_t \log \tilde{q}_t), \quad (57)$$

which gives [equations \(13\)](#) and [\(14\)](#).

A.3 Envelope derivative of the distortion

Write

$$D_t(\vartheta) = \lambda_t(\vartheta)^\top \mathbf{c}_t(\vartheta) - A_{t, \vartheta}(\lambda_t(\vartheta)). \quad (58)$$

Differentiation gives

$$\partial_{\vartheta} D_t = (\partial_{\vartheta} \lambda_t)^\top \mathbf{c}_t + \lambda_t^\top \partial_{\vartheta} \mathbf{c}_t - \partial_{\vartheta} A_{t, \vartheta}(\lambda_t) - \nabla_{\lambda} A_{t, \vartheta}(\lambda_t)^\top \partial_{\vartheta} \lambda_t \quad (59)$$

$$= \lambda_t^\top \partial_{\vartheta} \mathbf{c}_t - \partial_{\vartheta} A_{t, \vartheta}(\lambda_t), \quad (60)$$

using $\nabla_{\lambda} A = \mathbf{c}_t$. At fixed λ ,

$$\partial_{\vartheta} A_{t, \vartheta}(\lambda) = \mathbb{E}_{Q_t^{\text{fb}}}[\partial_{\vartheta} \log \tilde{q}_t], \quad (61)$$

which proves [equation \(17\)](#).

A.4 Minimal-energy dynamic correction

Let $\delta_t = v_t - u_t$. Since

$$\partial_t q_t + \nabla \cdot (q_t u_t) = q_t h_t, \quad (62)$$

v_t realizes q_t if and only if

$$\nabla \cdot (q_t \delta_t) = -q_t h_t. \quad (63)$$

If ψ_t solves [equation \(21\)](#), then $\delta_t^* = -\nabla \psi_t$ satisfies [equation \(63\)](#). For any other feasible correction write $\delta_t = -\nabla \psi_t + w_t$ with $\nabla \cdot (q_t w_t) = 0$. Weighted integration by parts gives

$$\mathbb{E}_{q_t}[\nabla \psi_t \cdot w_t] = 0, \quad (64)$$

so

$$\mathbb{E}_{q_t} \|\delta_t\|^2 = \mathbb{E}_{q_t} \|\nabla \psi_t\|^2 + \mathbb{E}_{q_t} \|w_t\|^2 \geq \mathbb{E}_{q_t} \|\nabla \psi_t\|^2. \quad (65)$$

This proves the minimality statement.

A.5 Eliminating the reference density derivative

Let $s_t = \nabla \log q_t$ and $\tilde{s}_t = \nabla \log \tilde{q}_t$. From [equation \(15\)](#),

$$s_t = \tilde{s}_t + J_{\Phi}^\top \lambda_t. \quad (66)$$

The reference continuity equation implies pointwise

$$\tilde{\ell}_t = -\nabla \cdot u_t - u_t^\top \tilde{s}_t. \quad (67)$$

Therefore

$$\tilde{\ell}_t = -(\nabla \cdot u_t + u_t^\top s_t) + \lambda_t^\top J_{\Phi} u_t. \quad (68)$$

For vector Φ , integration by parts under q_t yields

$$\mathbb{E}_{q_t}[(\Phi - \mathbf{c}_t)(\nabla \cdot u_t + u_t^\top s_t)] = -\mathbb{E}_{q_t}[J_{\Phi} u_t]. \quad (69)$$

Substitution gives [equation \(25\)](#) and hence [equation \(26\)](#).

Next, differentiating the log density in [equation \(9\)](#) gives

$$\partial_t \log q_t = \tilde{\ell}_t + \dot{\lambda}_t^\top (\Phi - \mathbf{c}_t) - \mathbb{E}_{q_t}[\tilde{\ell}_t]. \quad (70)$$

Consequently,

$$h_t = \partial_t \log q_t + \nabla \cdot u_t + u_t^\top s_t \quad (71)$$

$$= \dot{\lambda}_t^\top (\Phi - \mathbf{c}_t) - \mathbb{E}_{q_t}[\tilde{\ell}_t] + \lambda_t^\top J_\Phi u_t. \quad (72)$$

Finally, integration by parts with the constant test function implies

$$\mathbb{E}_{q_t}[\tilde{\ell}_t] = \lambda_t^\top \mathbb{E}_{q_t}[J_\Phi u_t], \quad (73)$$

which gives [equation \(27\)](#).

A.6 Deep-Ritz variational form

Consider

$$\mathcal{J}_t(\psi) = \frac{1}{2} \int q_t \|\nabla \psi\|^2 dx + \int q_t h_t \psi dx. \quad (74)$$

For any perturbation φ ,

$$\left. \frac{d}{d\epsilon} \mathcal{J}_t(\psi + \epsilon \varphi) \right|_{\epsilon=0} = \int q_t \nabla \psi \cdot \nabla \varphi dx + \int q_t h_t \varphi dx \quad (75)$$

$$= - \int \varphi \nabla \cdot (q_t \nabla \psi) dx + \int q_t h_t \varphi dx. \quad (76)$$

Stationarity for all test functions is equivalent to [equation \(21\)](#).

B Hackathon implementation checklist

Component	Minimal implementation	Pass criterion
Reference SI	MLP for 1D/2D; equivariant graph net for particles	Reproduces endpoint transport and low reference FM loss
Moment projection	Damped Newton on empirical dual with stabilized weights	Fresh weighted moments within tolerance; no silent rank failure
Implicit VJP	Custom linear solve through calibration	Central finite-difference agreement on toy parameter
$\dot{\lambda}$ and forcing	JVP $J_\Phi u$ plus weighted covariance statistics	Agreement with quadrature/finite differences in 1D
Poisson correction	Scalar network trained by weighted Deep-Ritz loss	Low continuity residual and low oracle path discrepancy
Generation	ODE with $u_\theta - \nabla \psi_\omega$	Endpoint fidelity plus flat measured moments across time
Safety layer	Optional ensemble Gram correction	Reduces empirical moment drift without large path distortion
Many-body bridge	$N \approx 32$, smooth pair moments, angular hidden metrics	Pair moments fixed while hidden structure changes materially

Table 1: A hackathon implementation can stop after the deterministic ODE. Score learning and the SDE are optional extensions.

C Suggested figure and table set for the final paper

1. **Figure 1: Concept and pipeline.** Reference stochastic interpolant leaves the moment fiber; I-projection bends the marginal path back; Poisson correction realizes it dynamically.

2. **Figure 2: Exact 1D validation.** Density path, moment error, λ_t , $\dot{\lambda}_t$, oracle versus learned correction velocity.
3. **Figure 3: 2D law-path comparison.** Raw SI, MGD, tangent-only, and MFSI snapshots with identical moment curves but different projected-path fidelity.
4. **Figure 4: Many-body fiber bridge.** Configurations along time, flat radial observables, changing angular/triplet observables.
5. **Figure 5: Compute and stability.** ESS, projection distortion, correction energy, NFE, and wall-clock cost versus reference coupling/schedule.
6. **Table 1: Related work.** Constraint object, target object, whether full law path is specified, whether constraints are population-level, and whether transport is within an observational equivalence class.
7. **Table 2: Quantitative results.** Moment error, projected-path discrepancy, endpoint discrepancy, hidden-structure coverage, and compute.

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